

Hotel Cancellation Project

This project uses the Hotel Reservations Classification Dataset available on Kaggle.

Dataset source:

Quantitative Data

- <https://www.kaggle.com/datasets/ahsan81/hotel-reservations-classification-dataset>
- The dataset contains hotel booking information including guest characteristics, booking details, room pricing, and booking outcomes. The target variable used in this project was booking_status, indicating whether a booking was cancelled or not.

Qualitative Data

- Industry questionnaire responses from hospitality professionals

Hypothesis

H1: Lead time affects cancellations

Bookings made far in advance are more likely to cancel

H2: Price affects cancellations

Higher-priced bookings are more likely to cancel

H3: Previous cancellations impact

Previous cancellations are more likely to cancel again

H4: Repeated guests cancel less

Returning customers are more reliable

H5: Special requests indicate commitment

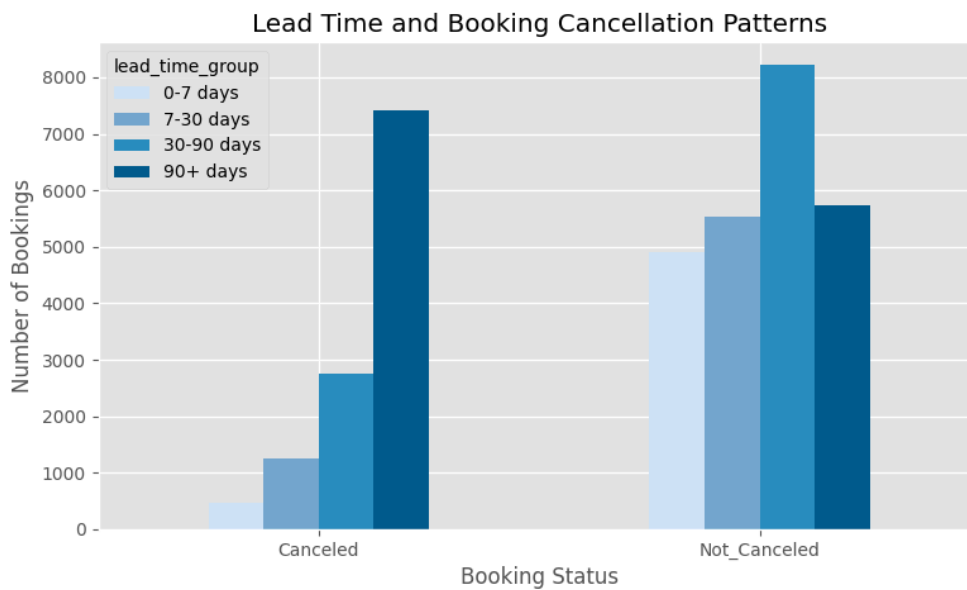
Guests with more special requests are less likely to cancel

H6: Booking composition matters

Families / larger groups behave differently from solo travellers

1st Query Lead time vs Cancellation

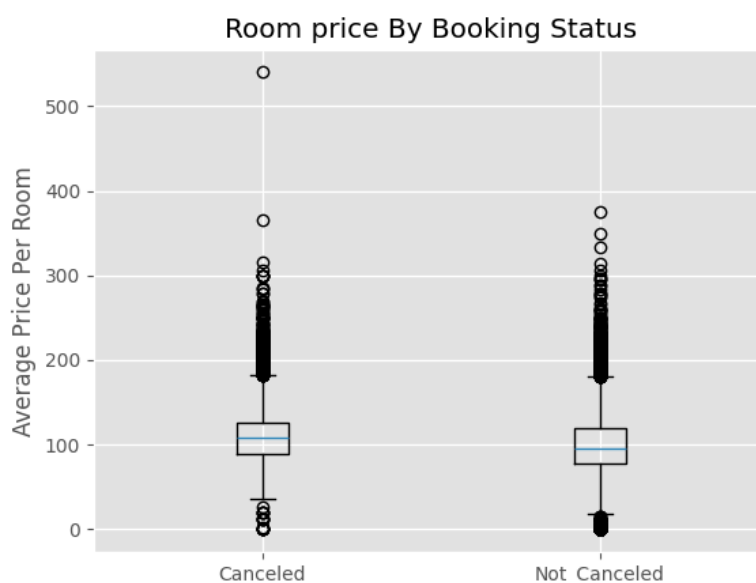
- The grouped bar chart shows a large increase in cancellations the longer the lead time towards bookings is, especially a substantial increase after 90+ days. Suggesting an increase in booking uncertainty over longer periods of time, indicating that lead time could be a strong predictive feature for cancellation risk.
- **Multivariate graphical EDA**



2nd Query Booking status vs Average booking price

- Although cancelled bookings show a slightly higher average price than non cancelled bookings, as indicated by the box plot between the two groups the distributions overlap substantially and the median values are similar. Indicating that room price alone is a weak standalone predictor of cancellations.

Booking_status	Avg_price
Canceled	110.59
Not Canceled	99.93



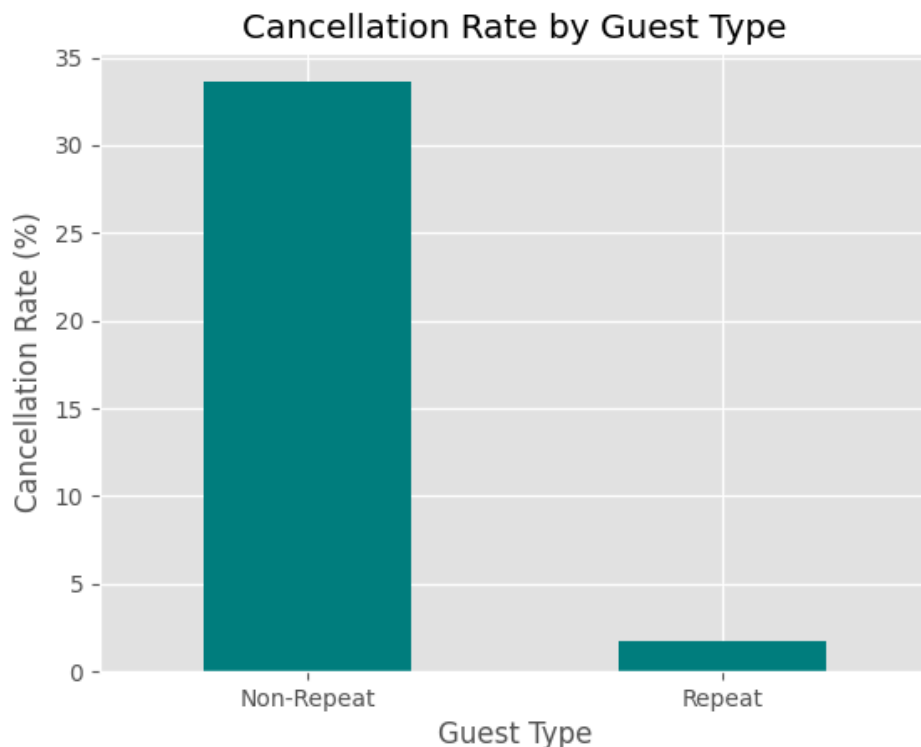
3rd Query Impact of Previous Cancellations

- Although guests with higher prior cancellation history showed slightly higher cancellation rates than those with low cancellation history, the sample sizes were substantially smaller. Industry insight may indicate operational factors such as stricter cancellation policies or reduced repeat bookings after multiple cancellations could contribute to the rarity of high cancellation history guests. Therefore previous cancellation history alone would be a weak predictor of cancellations.

cancellation_history	cancelled_bookings	total_bookings	cancellation_rate
None	11869	35937	33.03
Low (1-2)	11	244	4.51
High (3+)	5	94	5.32

4th Query Repeated Guests cancel less

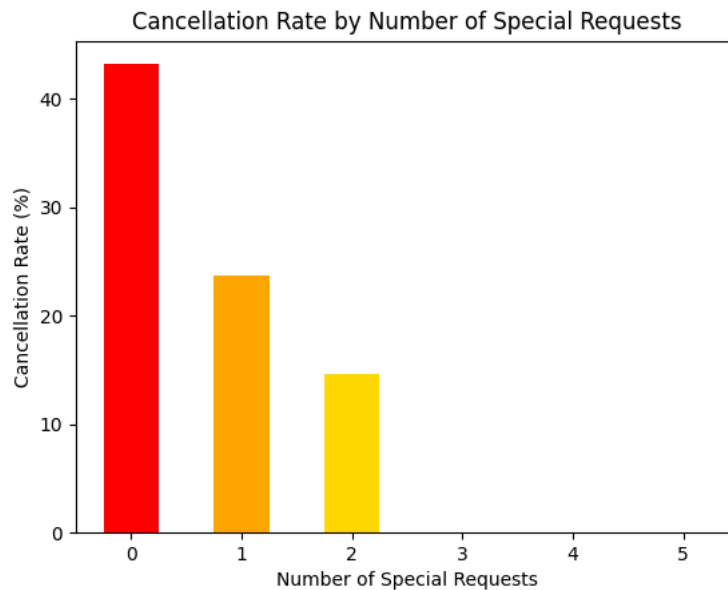
- Repeated guests have demonstrated significantly lower cancellations rates, in comparison to non repeating guests. Suggesting that customer loyalty and prior booking history, may reduce the likelihood of cancellation. Making repeated guest status a valuable predictive feature.



Query 5 Special requests indicate commitment

- Cancellation rates decreased substantially as special requests increased, although categories with 3 or more requests contained smaller sample sizes. As special

requests increase, cancellation rate decreases , indicating that special requests have a strong predictive feature.



Query 6 Booking composition matters

- Group size appears to demonstrate a moderate relationship with cancellation behaviour, although the effect is weaker than features such as lead time, repeated guest status, and special requests

Cancellation Rates by Group Size			
total_guests	cancelled_bookings	total_bookings	cancellation_rate
1	1809	7552	23.95%
2	8280	23942	34.58%
3	1392	3851	36.15%
4	398	912	43.64%
5	5	15	33.33%

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Feature

Likely predictive strength

- Lead time
- Repeated guest

Strong

Very strong

- Special requests **Very strong**
- Group size **Moderate**
- Previous cancellations **Moderate / weak**
- Average room price **Weak**

```
0.774776016540317
```

	precision	recall	f1-score	support
0	0.79	0.90	0.84	4839
1	0.73	0.52	0.60	2416
accuracy			0.77	7255
macro avg	0.76	0.71	0.72	7255
weighted avg	0.77	0.77	0.76	7255
[[4376 463]				
[1171 1245]]				

- Accuracy 77.47 %
- The model correctly predicted around 77/100 bookings.

	Feature	Coefficient
0	lead_time	1.075944
4	avg_price_per_room	0.618694
3	total_guests	0.099219
5	no_of_previous_cancellations	0.097141
1	repeated_guest	-0.448474
2	no_of_special_requests	-0.852109

Feature	Coefficient	
lead_time	+1.076	Strongly increases cancellation probability
avg_price_per_room	+0.619	Higher prices increase cancellation probability
total_guests	+0.099	Small increase
previous_cancellations	+0.097	Very weak positive effect

repeated_guest	-0.448	Reduces cancellation probability
special_requests	-0.852	Strongly reduces cancellation probability

Actual	Predicted Not Cancelled	Predicted Cancelled
Not Cancelled	4376	463
Cancelled	1171	1245

Metric	Result
Accuracy	77.48%
Recall (Cancelled)	52%
Precision (Cancelled)	73%
False negatives	1171

Evaluation :

Logistic regression results are aligned largely with exploratory analysis. With lead time showing the strongest positive correlation with cancellations. Whereas repeated guest status and number of special requests demonstrated strong negative relationships. Another interesting observation showed that average price demonstrated more strongly than expected than EDA findings alone would suggest, highlighting how features can be more informative than when considered together.

Comparison with Industry Insight :

Preliminary feedback from hospitality professionals broadly aligned with model findings. Respondents identified booking lead time and first-time guests as key contributors to cancellations, which corresponded with logistic regression results showing lead time as the strongest positive predictor and repeat guest status as a negative predictor. Respondents also suggested that changes in travel plans, emergencies, and the availability of cheaper alternatives were common reasons for cancellations, providing possible real-world explanations for the observed relationship between longer lead times and increased cancellation likelihood.

Refining model :

The initial model achieved 77% accuracy but missed a substantial number of cancellations. A second model using balanced class weights was tested to improve recall and reduce false negatives.

```
0.7547898001378359
      precision    recall  f1-score   support

     0       0.85      0.77      0.81      4839
     1       0.61      0.73      0.66      2416

 accuracy      0.75      7255
 macro avg      0.73      0.75      0.74      7255
weighted avg      0.77      0.75      0.76      7255

[[3721 1118]
 [ 661 1755]]

      Feature    Coefficient
0      lead_time      1.100191
4  avg_price_per_room      0.599996
5 no_of_previous_cancellations      0.092497
3      total_guests      0.082347
1      repeated_guest     -0.467382
2  no_of_special_requests     -0.858358
```

Metric	Value
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Accuracy	75.5%
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Precision (Cancelled)	61%
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Recall (Cancelled)	73%
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False Negatives	661
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Coefficients

Lead Time	1.100
Special Requests	-0.858
Repeated Guest	-0.467

Underlying relationships are stable. The balanced model sacrificed a minimal amount of accuracy, but improved in a substantial reduction in missed calculations, improving the recall from 52% to 73%.

Conclusion :

This project demonstrated that hotel booking cancellations could be predicted with a fair degree of accuracy based on booking and guest information. The exploratory analysis showed that lead time, repeat guest status and special requests were the strongest predictors of cancellation behaviour. These findings were largely supported by logistic regression results, with lead time being the strongest positive predictor and repeat guest status and special requests being strong negative predictors.

A balanced logistic regression model improved cancellation detection by increasing recall from 52% to 73%, reducing missed cancellations while similar underlying feature relationships were maintained. Industry feedback was mostly consistent with the quantitative results, and offered more operational context for the observed cancellation patterns.

Limitations :

- The dataset was obtained from Kaggle and may not fully represent the behaviour of a specific hotel or market segment.
- Industry questionnaire responses were limited and should be considered exploratory rather than statistically representative.
- Additional features such as weather, flight disruptions, economic conditions, or competitor pricing were not available within the dataset.
- The model predicts cancellation likelihood but does not identify the specific reason for cancellation.